Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Agentic Systems · Evaluation Pipelines · Document Parsing · LLMs in Production · Python · AWS · FastAPI

venkatalakshmiparimalap@gmail.com | +1-703-261-5410 | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti |
AgentProbe (eval toolkit) | Virginia · Open to Remote / Hybrid

PROFESSIONAL SUMMARY

I build **agentic systems that automate high-stakes decisions** — from document parsing pipelines to autonomous agent workflows with production eval harnesses. 7+ years of backend engineering experience, now focused on shipping AI systems that work in production. Built **Leasel**: a document parsing + agentic investigation pipeline (8-agent LangGraph StateGraph, OCR ingestion, section-aware chunking, hybrid retrieval, 93.6% precision across 1,240 test cases). Built **AgentProbe**: an open-source **evaluation pipeline** for AI/ML systems (5-metric LLM-as-judge engine, CI-gated regression detection, dynamic routing across 5 LLM providers). Currently at Intuit integrating Anthropic Claude into financial applications — **pydantic-validated structured outputs**, RAG grounding, eval harness catching regressions before release. Python (9+ yrs), **AWS Lambda / RDS / S3 / PostgreSQL**, FastAPI, LangGraph, LangChain. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). — github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Document Parsing Pipelines | Agentic Systems & Workflows | Evaluation Pipeline Design | LLM-as-Judge & Eval CI | Audit Automation | Rules-Based to Autonomous Systems | LLMs in Production | RAG Systems & Hybrid Retrieval | Python & pydantic | AWS Lambda / RDS / S3 | Multi-Agent Architectures | Production AI Guardrails

EXPERIENCE

Intuit

Jun 2026 – Present · Virginia, USA

Software Engineer — LLMs in Production · Evaluation Pipelines · Python · pydantic · AWS

- Integrate Anthropic Claude into Python/FastAPI financial services: **pydantic-validated structured output schemas** constrain model responses for reliable downstream parsing and audit-trail generation; RAG grounding traces every Claude response to source financial records.
- Build and own the **evaluation pipeline**: LLM-as-judge scoring with golden datasets and CI regression detection; caught 3 prompt regressions before release; zero hallucination incidents post-deployment; prompt versioning enables A/B testing across model versions.
- Design **agentic workflow guardrails**: input validation, sensitive-data redaction, output schema enforcement, fallback paths for failed LLM calls; containerize services with Docker and deploy through GitHub Actions CI/CD to **AWS** (Lambda, S3, RDS) with OpenTelemetry monitoring.
- Build ML pipelines (scikit-learn, XGBoost) from feature engineering to prediction APIs deployed on AWS; implement React/TypeScript front ends consuming FastAPI backends.

Entrata (Apartment & Affordable Housing SaaS)

Oct 2020 – Jul 2024 · 100K+ MAU, 1M+ property records

Senior Software Engineer — Python · Java · AWS S3/RDS · PostgreSQL · Document Pipelines

- Shipped 20+ production features end-to-end (Python, Java/Spring Boot, PHP, PostgreSQL, React) across 120+ Agile sprints; **document parsing and audit automation**: income verification document pipelines processing 1.6M HUD/Tax Credit records with validation, rules enforcement, and compliance checks.
- Built Python/Bash automation eliminating 80% of manual operational work; internal debugging and data investigation tools cutting triage time 40%; optimized 200+ SQL queries (30% performance improvement via execution plan analysis and indexing).
- **AWS S3 PHI migration** with Libsodium encryption (100% PHI compliance, zero incidents); Docker containerization cutting microservices execution time 50%; **PostgreSQL / RDS** schema design and optimization across multi-tenant platform.
- RabbitMQ async processing pipeline for bulk tenant record updates with idempotency guarantees; reviewed 300+ PRs; mentored junior engineers.

Entrata India (Db Xento Systems)

Sep 2018 – Oct 2020 · Pune, India

Junior Software Engineer — Python · PHP · Java · PostgreSQL · AWS

- Built property management modules and **compliance document pipelines** processing 1.6M HUD/Tax Credit records with rules-based validation; led AWS S3 cloud migration from legacy storage; created 56 state-specific housing certification documents.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

AI/ML Research · Fine-Tuning · PyTorch · HuggingFace · Statistical Modeling

- Fine-tuned classification model with HuggingFace Transformers + PyTorch (data prep → training → eval → deploy on Quart), presented to GMU faculty; built predictive models (R^2 = 0.977, 190+ countries); applied Isolation Forest anomaly detection. Mentored 600+ students.

PORTFOLIO PROJECTS

Leasel — Document Parsing + Agentic Investigation Pipeline

Document Parsing · Agentic Pipeline · Eval Harness

2026 · Python, LangGraph (8-agent StateGraph), FastAPI, OCR, TF-IDF + semantic hybrid retrieval, SQLite, Anthropic/OpenAI APIs · github.com/venkatalakshmiparimala/Lease_complaince_rag

- **Document parsing pipeline**: OCR ingestion → section-aware chunking → hybrid TF-IDF + semantic reranking retrieval → answers cited to exact source clause; 100% citation correctness, 0% unsupported-claim rate on eval set.
- **8-agent LangGraph agentic investigation pipeline**: triage → retrieval → Config/History/StructuredData parallel agents → reasoning → validation → confidence-gated HITL escalation; correctly traces false-ineligibility ticket to misconfigured setting tagged to specific release — exact audit automation pattern.
- **Evaluation pipeline** via AgentProbe LLM-as-judge CI harness: 93.6% precision, 87.9% F1 across 1,240 test cases; **audit rules enforced in code**: LLM reasons, deterministic validator decides every regulated call; safety boundary adversarially tested.

AgentProbe — Open-Source Evaluation Pipeline for AI/ML Systems

Evaluation Pipeline · LLM-as-Judge · Agentic Systems

2025 · Python, FastAPI (15 endpoints), LangGraph (3 agents), ChromaDB, Anthropic, OpenAI, Azure OpenAI, Groq, Ollama · github.com/venkatalakshmiparimala/AgentProbe

- **Evaluation pipeline**: 5-metric LLM-as-judge engine (factual accuracy, hallucination rate, retrieval quality, completeness, compliance); CI-integrated regression detection blocks production deploys when quality drops; used to evaluate Leasel across 1,240 test cases.
- 3-agent LangGraph agentic system (Orchestrator, RAG Tracer, Auto Fix) with multi-step planning; LLM Router for dynamic routing across 5 LLM providers; 15-endpoint FastAPI backend; 100% local at zero cost — measures and monitors model/agent performance in production-equivalent conditions.

AI on FHIR — Clinical NLP & Document Query Automation

Document Automation · LLMs in Production

2025 · Python, Flask, React, TypeScript, Docker, BERT, spaCy, LLMs, FHIR REST APIs · github.com/venkatalakshmiparimala/AI_on_FHIR

- NLP + LLM pipeline translating English/Spanish clinical questions into structured FHIR API queries; HIPAA guardrails (RBAC, PHI output filtering); ~60% reduction in manual document lookup time; containerized with Docker.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

AI & Agentic Systems: Agentic Systems, Agentic Pipelines, Document Parsing Pipelines, Evaluation Pipelines, Audit Automation, LLMs in Production, LLM-as-Judge, Eval CI/CD, Rules-Based to Autonomous Workflows, Multi-Agent Architectures, LangGraph, LangChain, LangGraph StateGraph, RAG Systems, Hybrid Retrieval, Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama, pydantic, Structured Outputs, Human-in-the-Loop (HITL), Guardrails, Hallucination Detection, Fallback Paths, OCR, Embeddings, ChromaDB, Semantic Search

Engineering & Backend: Python (9+ yrs), FastAPI, Flask, Django, Java, Spring Boot, TypeScript, Node.js, React, PHP, SQL, PL/SQL, Bash, PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, Snowflake, ETL Pipelines, Data Modeling, Query Optimization

Cloud & DevOps: AWS (Lambda, RDS, S3, EC2, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, Databricks, AKS), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, OpenTelemetry, Prometheus, Grafana, RabbitMQ, Kafka, Celery

ML & Data: scikit-learn, XGBoost, PyTorch, HuggingFace Transformers, Fine-Tuning, BERT, spaCy, Isolation Forest, Pinecone, Weaviate, Elasticsearch, TF-IDF, Vector Search, Pandas, NumPy, PySpark